Giorgi Malania

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EDUCATION

Case Western Reserve University · Cleveland, OH **May 2028**

B.S. Mechanical & Aerospace Engineering · GPA: 3.8/4.0 · Dean's High Honor List

- Study Abroad: Universidad Carlos III de Madrid (UC3M), Fall 2026
- Coursework: Heat Transfer, Strength of Materials, Fluid Mechanics, Thermodynamics, Dynamics, Mechanical Manufacturing, Circuits

TECHNICAL SKILLS

Design: SolidWorks (CSWA Certified), Engineering Drawings, GD&T (ASME Y14.5)

Simulation & Analysis: SolidWorks Simulation (FEA), Vortex Panel Method, Numerical Integration

Test & Instrumentation: LabVIEW, Data Acquisition (DAQ), Sensor Fault Isolation, NI cDA

Manufacturing: Manual Lathe & Mill, FDM 3D Printing, CNC Machining, Fusion 360 CAM

Software: Python (NumPy, SciPy, Matplotlib), MATLAB, Jupyter, TypeScript, JavaScript, Next.js, React, Node.js, PostgreSQL, Supabase, REST APIs, OAuth, Git, LaTeX/TikZ, LLM Pipeline Design

PROJECTS

Hurdle Helper · Regulation Hurdle Training Attachment Sep 2025 – Dec 2025

- Achieved a 1.8× yield safety factor on AL 6061-T6 by iterating wall thickness and fillet geometry against FEA across 3 prototype rounds, holding peak bending stress to 19.1 ksi against a 35 ksi yield
- Eliminated manual reset between runs with a spring-loaded return mechanism on a ball-bearing pivot
- Enabled fabrication without designer input by releasing a 4-sheet drawing package to ±.005" tolerances

NACA 2415 Wind Tunnel Characterization Apr 2026

- Measured a peak lift coefficient of 1.09 at 10° and located stall onset by acquiring 16-tap pressure data through LabVIEW at ∼20 m/s across seven angles (−10° to 20°)
- Resolved lift and moment coefficients to within 15–25% of vortex panel predictions by integrating C_p in Python
- Kept all seven sweeps usable by correcting a faulty tap via neighbor averaging and excluding a second from integration

Precision-Machined Steel Hammer Sep 2025 – Dec 2025

- Held ±.005" across mating features by machining a two-component hammer from 1018 CRS on manual lathe and mill
- Assembled two components to drawing spec by cutting 5/16-18 UNC threads and reaming Ø.438/.439" bores

Oushi · Personal AI Assistant May 2026

- Cut LLM ranking calls by 50–70% by writing a regex prefilter that scores obvious noise before model invocation
- Shipped to production at oushi.app a real-time Next.js/TypeScript pipeline that ingests Gmail over Google Pub/Sub push, classifies with the Anthropic API, and writes priority labels back to the inbox

PROFESSIONAL EXPERIENCE

General Learning (YC F24) · *Hong Kong* May 2025 – Aug 2025 · Jun 2026 – Present

Content Product Manager

- Cut technical diagram production time by over 50% by designing a prompt-engineered LLM pipeline that classifies exam questions by problem type and compiles TikZ/Desmos/SVG to print-ready figures
- Led production of 5,000+ original questions and 4,000+ instructional videos across physics, mathematics, and standardized-test material for platforms serving ∼1M students
- Authored physics textbooks and question banks for both B2B classroom and B2C direct-to-student use

Case Western Reserve University · Tinkham Veale University Center Jan 2025 – Jun 2026

Special Events Assistant

- Delivered 4–5 campus events weekly by coordinating setup, logistics, and equipment for up to 500 attendees
- Turned over event spaces between back-to-back bookings by running setup, breakdown, and front-desk operations

Weatherhead School of Management · Cleveland, OH Dec 2024 – Jun 2026

Executive Assistant

- Ran registration and logistics for daily executive cohorts of 30+ from JPMorgan, KeyBank, and Lockheed Martin
- Consolidated the business development pipeline by building the school's outreach database of 1,500+ contacts from scratch

LEADERSHIP & HONORS

YQuantum 2026 Hackathon · *Yale University* · 2nd Place of 35 Teams Apr 2026

- Placed 2nd of 35 by compressing a 50-asset insurance portfolio onto 8 qubits as a constrained mean-variance QUBO and solving it on QuEra's Bloqade neutral-atom simulator
- Cut circuit depth up to 48% and raised simulated hardware fidelity from 17% to 53% under Gemini noise by implementing QAOA with K_8 edge-coloring parallelization

WSOM Business Case Competition · *2nd Place of 16 Teams, $500 Award* Mar 2026

- Placed 2nd of 16 by building a customer, DTC, and growth strategy for a $140M manufacturer in one hour

Consult Your Community · *Business Analyst, Haven MVP launch* Jan 2026 – Jun 2026